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Abstract

In this work we characterize the identification capacity region of a general broadcast channel. The direct part is based on code construction which makes use of the common randomness inherent to any transmission coding scheme with positive common message rate. The converse part makes use of a distribution packing lemma, applied to each of the message sets separately.

Index terms — Broadcast channel, common experiment, degraded message sets, distribution packing, identification via channels.

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1 Introduction

The problem of identification via channels was first introduced by Ahlswede and Dueck in [2], [3]. In the identification problem, first formulated for single user channels, the receiver's task is to decide whether one specific message, whose index is not predetrmined, was transmitted or not. This task is easier than the task of the receiver in the classical Shannon's transmission system, where the receiver is required to decide which message was sent. Ahlswede and Dueck showed in [2] that if stochastic encoders are used, the number of messages which can be identified grow doubly exponentially with blocklength. Therefore, the identification (ID) capacity is defined as the normalized iterated logarithm of the number of messages that can be identified via the channel. In their original paper, Ahlswede and Dueck showed that the ID capacity of a single user channel is lower bounded by the Shannon capacity. A regular converse result was not proved in the original paper. Instead, it was shown that if the probabilities of error are required to vanish exponentially fast with blocklength, then ID capacity equals the Shannon capacity of the channel. They have termed this result as *soft converse*. A converse with fixed probabilities of error was proved by Han and Verdú in [7].

In [10], Verboven and van der Meulen studied the problem of identification via the broadcast channel (BC). In particular, they obtained an inner bound on the ID capacity region of a BC with feedback (BCF), and an outer bound on the ID capacity region of BC without feedback, under the requirement of exponentially small probabilities of error. Following the terminology of [2], we denote such a result as *soft outer bound*. The inner bound for the BCF exceeds, in general, the soft outer bound for the BC, thus the authors conclude that feedback increases the ID capacity of the BC. Moreover, for the deterministic BC, the two bounds coincide, thus yielding a characterization of the ID capacity region for that channel. (Note that any deterministic channel inherently contains a feedback, and therefore the inner bound is relevant.) An important point raised by the authors about this result is that it exceeds the transmission capacity region of the deterministic broadcast channel, and therefore in multiterminal systems, ID capacity region can exceed transmission capacity region.

In this work we characterize the ID capacity region of the general discrete broadcast channel. Our framework differs from that of [10] in two main aspects. Our model does not assume feedback,

as is assumed in the inner bounds of [10]. In addition, the outer bounds do not require the probabilities of error to vanish exponentially fast with blocklength. I.e., we derive regular – rather than soft – outer bounds. The inner bounds are derived based on the concept of *common experiment*, introduced in [3]. The outer bound is based on a distribution packing result [9], which was derived initially to study the ID capacity region of the multiple access channel, and is used here for the BC.

Definition of the ID problem for the broadcast channel, and some known results concerning transmission via that channel, are given in Section 2. The results on transmission are relevant as the construction of ID code used in the proof of the direct part is based on transmission codes. Our main result is stated and proved in Section 3.

2 Definitions

Let $\mathcal{X}$, $\mathcal{Y}$, $\mathcal{Z}$ be finite sets. A broadcast channel $(\mathcal{X}, W(y,z|x), \mathcal{Y} \times \mathcal{Z})$ is a channel with input alphabet $\mathcal{X}$, output alphabet $\mathcal{Y} \times \mathcal{Z}$, and transition probability matrix $W(y,z|x)$. When the choice of the alphabets is clear, we will refer to the broadcast channel by $W(y,z,|x)$. We denote the marginals by W_y and W_z, i.e.,

$$\begin{aligned} W_y(y|x) &= \sum_z W(y,z|x), \\ W_z(z|x) &= \sum_y W(y,z|x). \end{aligned}$$

The channel is memoryless if for n-sequences the transition behaves according to the probabilities

$$W^n(y^n, z^n|x^n) = \prod_{i=1}^{n} W(y_i, z_i|x_i).$$

A broadcast channel is said to be *degraded* if the triplet $X - Y - Z$ form a Markov chain for any X, i.e., if there exists some transition probability $W'(z|y)$ such that

$$W(y,z|x) = W_y(y|x)W'(z|y),$$

in which case we say that Z is the degraded component. In the sequel, the output distribution of a channel W^n when the input distribution is Q will be denoted as $QW^n(\cdot)$, i.e.,

$$QW^n(y^n, z^n) = \sum_{x^n} W^n(y^n, z^n|x^n)Q(x^n),$$

and similar notation holds for the marginl channels.

Definition 1 An $(n, N_y, N_z, \lambda_1, \lambda_2)$ *ID code for the broadcast channel* $W^n(y^n, z^n|x^n)$ is a collection of triplets $\{(Q_{i,j}, \mathcal{B}_i^y, \mathcal{B}_j^z),\ i = 1, 2, \ldots N_y,\ j = 1, 2, \ldots N_z\}$, such that

1. $Q_{i,j}$ is a distribution on $\mathcal{X}^n$
2. $\mathcal{B}_i^y \subset \mathcal{Y}^n$, and $\mathcal{B}_j^z \subset \mathcal{Z}^n$, for all $i = 1, 2, \ldots N_y$, $j = 1, 2, \ldots N_z$.
3. $Q_{i,j}W^n(\mathcal{B}_i^y \times \mathcal{B}_j^z) \geq 1 - \lambda_1, \quad \forall\ i, j$
4. $Q_{i,j}W^n(\mathcal{B}_k^y \times \mathcal{B}_\ell^z) \leq \lambda_2, \quad \forall\ (k, \ell) \neq (i, j)$.

The *rate pair* of the code (R_y, R_z) is the normalized iterated logarithm of the code size

$$(R_y, R_z) = \frac{1}{n}(\log\log N_y, \log\log N_z).$$

A rate pair (R_y, R_z) is called (λ_1, λ_2)-*achievable ID rate* for the broadcast channel W if for every $\gamma > 0$ and sufficiently large n there exists an $(n, N_y, N_z, \lambda_1, \lambda_2)$ ID code for W^n, with rate pair

$$\begin{aligned} \frac{1}{n}\log\log N_y &\geq R_y - \gamma, \\ \frac{1}{n}\log\log N_z &\geq R_z - \gamma. \end{aligned}$$

The closure of all (λ_1, λ_2)-achievable ID rate pairs is called the (λ_1, λ_2)-ID capacity region, and is denoted by $\mathcal{C}_{ID}(\lambda_1, \lambda_2)$. The identification capacity region, $\mathcal{C}_{ID}$, is

$$\mathcal{C}_{ID} = \bigcap_{\lambda_1, \lambda_2 > 0} \mathcal{C}_{ID}(\lambda_1, \lambda_2).$$

One can verify that, as is the case with transmission, the ID capacity region of the broadcast channel depends only on the marginal channels W_y and W_z. For example, onditions 3 and 4 of Definition 1 imply that

$$Q_{i,j}W_y^n(\mathcal{B}_i^y) \geq 1 - \lambda_1 \tag{1a}$$

$$Q_{i,j}W_z^n(\mathcal{B}_j^z) \geq 1 - \lambda_1, \tag{1b}$$

and

$$Q_{k,\ell}W_y^n(\mathcal{B}_i^y) \leq \lambda_1 + \lambda_2, \quad \forall \ell \text{ and } i \neq k \tag{2a}$$

$$Q_{k,\ell}W_z^n(\mathcal{B}_j^z) \leq \lambda_1 + \lambda_2, \quad \forall k \text{ and } j \neq \ell. \tag{2b}$$

Therefore conditions 3 and 4 of Definition 1 guarantee good ID codes for each of the marginals separately. Similarly, using equations (1) and (2) imply good ID code in the sense of Definition 1, although with slightly higher probabilities of error.

By Definition 1, the output distributions of a good ID code are pairwise separated in the variational distance, i.e.,

$$\begin{aligned} d(Q_{i,j}W^n, Q_{k,\ell}W^n) &\stackrel{\Delta}{=} 2 \max_{\mathcal{A}\subset\mathcal{Y}^n\times\mathcal{Z}^n} (Q_{i,j}W^n(\mathcal{A}) - Q_{k,\ell}W^n(\mathcal{A})) \\ &\geq 2Q_{i,j}W^n(\mathcal{B}_i^y \times \mathcal{B}_j^z) - 2Q_{k,\ell}W^n(\mathcal{B}_i^y \times \mathcal{B}_j^z) \\ &\geq 2(1-\lambda_1-\lambda_2), \quad \forall (i,j) \neq (k,\ell). \end{aligned} \tag{3}$$

This can be specialized to each of the marginal channels

$$d(Q_{i,j}W_y^n, Q_{k,\ell}W_y^n) \geq 2(1-\lambda_1-\lambda_2) \quad \forall \quad i \neq k, \tag{4}$$

$$d(Q_{i,j}W_z^n, Q_{k,\ell}W_z^n) \geq 2(1-\lambda_1-\lambda_2) \quad \forall \quad j \neq \ell. \tag{5}$$

In proof of the direct part of our main result we make use of a transmission code for the broadcast channel. We thus recall the relevant definitions and known results.

Definition 2 An $(n, M_y, M_z, M_0, \lambda)$ transmission code for the broadcast channel W^n is a collection of codewords

$$\{\boldsymbol{x}_{i,j,k} \in \mathcal{X}^n, \quad 1 \leq i \leq M_y, \quad 1 \leq j \leq M_z, \quad 1 \leq k \leq M_0\}$$

and corresponding decoding sets

$$\left\{\mathcal{D}_{i,k}^y \in \mathcal{Y}^n, \quad \mathcal{D}_{j,k}^z \in \mathcal{Z}^n, \quad 1 \leq i \leq M_y, \quad 1 \leq j \leq M_z, \quad 1 \leq k \leq M_0\right\}$$

such that $\{\mathcal{D}_{i,k}^y\}$ and $\{\mathcal{D}_{j,k}^z\}$ are disjoint collections, and

$$W_y^n(\mathcal{D}_{i,k}^y|\boldsymbol{x}_{i,j,k}) \geq 1-\lambda, \quad W_z^n(\mathcal{D}_{i,k}^z|\boldsymbol{x}_{i,j,k}) \geq 1-\lambda, \quad \forall\, i,j,k.$$

The rate triplet of the code is

$$(R_y, R_z, R_0) = \frac{1}{n}(\log M_y, \log M_z, \log M_0).$$

For convenience, we use the notation $\mathcal{M}$ to denote a transmission index set, i.e.,

$$\mathcal{M}_y = \{1, 2, \ldots, M_y\} \tag{6}$$

and similarly for $\mathcal{M}_z$ and $\mathcal{M}_0$, and we denote the encoder map by f

$$f: \quad (i,j,k) \longmapsto x_{i,j,k}, \quad (i,j,k) \in \mathcal{M}_y \times \mathcal{M}_z \times \mathcal{M}_0. \tag{7}$$

A triplet (R_y, R_z, R_0) is called λ-*achievable* transmission rate for W if, for any $\gamma > 0$ and sufficintly large n there exists an $(n, M_y, M_z, M_0, \lambda)$ transmission code for W^n with rates

$$\begin{aligned}
\frac{1}{n}\log M_y &\geq R_y - \gamma, \\
\frac{1}{n}\log M_z &\geq R_z - \gamma, \\
\frac{1}{n}\log M_0 &\geq R_0 - \gamma.
\end{aligned}$$

The determination of the set of all achievable triplets (R_y, R_z, R_0) for the general broadcast channel is still unknown, except for special models and cases. A model of independent interest is the broadcast channel with degraded message sets, suggested and solved by Körner and Marton in [8]. The broadcast channel with degraded message sets corresponds to the case when either of the rates R_y and R_z is 0. This is the asymmetric broadcast channel, ABC, in the terminology of [6]. To specify a code for the asymmetric broadcast channel, we will use the notation of Definition 2, ignoring the indices corresponding to the zero-rate component. Thus an (n, M_y, M_0, λ) code is a code for the ABC where the Z component gets only common messages, and the decoding sets are denoted by $\{\mathcal{D}^y_{i,k}\}$ and $\{\mathcal{D}^z_k\}$. Similar convention is used for the case where Y gets only the common messages. We state here the relevant result, which will be used in the next section. The transmission capacity region of the broadcast channel W with degraded message sets, denoted here $\mathcal{C}_{a,TR}$, is given by

$$\begin{aligned}
\mathcal{C}_{a,TR} = \bigcup_{P_U P_{X|U}} \{(R_y, R_0): \quad R_y + R_0 &\leq I(X;Y|U) + I(U;Z) \bigwedge I(X;Y), \\
R_0 &\leq I(U;Z)\},
\end{aligned} \tag{8}$$

where U is an arbitrary random variable, taking values in a finite set $\mathcal{U}$, whose cardinality can be bounded to satisfy $\|\mathcal{U}\| \leq \|\mathcal{X}\| + 2$. In the proof of our main result we will make use of transmission codes with degraded message sets. As a final remark, throughout the random variables will occasionally be replaced by their distributions, i.e., the mutual information will be denoted by $I(X;Y)$ and $I(P_X; P_X W)$ interchangeably.

3 Main Results

The next theorem states our main result.

Theorem 1 The identification capacity region of a discrete, memoryless broadcast channel is given by

$$\mathcal{C}_{ID} = \bigcup_{X} \{(R_y, R_z): \quad R_y \leq I(X;Y), \quad R_z \leq I(X;Z) \ \}. \tag{9}$$

The ID capacity region exceeds the transmission capacity region of the channel. In addition, note that it is convex, although time sharing cannot yield convex combinations of rates, due to the iterated logarithm.

Note that the expression for the ID capacity region given in Theorem 1 indicates that, for an input distribution X, identification for one user does not interfere with identification for the other user. This is in contrast to the situation in transmission. Recall that the transmission capacity region of the degraded broadcast channel is given by ([1], [4], [5])

$$\begin{aligned} \mathcal{C}_{\text{degraded},TR} = \bigcup_{P_U P_{X|U}} \{(R_y, R_z, R_0): \quad R_z + R_0 &\leq I(U;Z), \\ R_y &\leq I(X;Y|U)\}, \end{aligned} \tag{10}$$

where U is as in (8). Let us examine (10) for $R_0 = 0$. For a fixed input distribution X, the dependence of U on X determines the rate allocated to the degraded user Z, and accordingly, the reduction in the rate of Y due to the transmission to Z. The amount of loss to R_y is $I(X;Y) - I(X;Y|U) = I(U;Y)$, which, by the data processing inequality, is lower bounded by $I(U;Z)$. In other words, for any input distribution X, the rate loss of user Y due to the transmission to Z, equals at least the rate of transmission to Z. As seen by (9), this is not the situation in identification: although both channels share the same input, each channel performs the best that can be expected with input distribution X, regardless of the presence of the other channel.

The proof of Theorem 1 is based on a sequence of lemmas. We start with the direct part. For a given pair of random variables (U, X) with joint distribution $P_U P_{X|U}$, define the region

$$\begin{aligned} \mathcal{R}_i^y(U,X) \stackrel{\Delta}{=} \{(R_y, R_z): \quad R_y &\leq I(X;Y|U) + I(U;Z) \bigwedge I(X;Y), \\ R_z &\leq I(U;Z)\}. \end{aligned} \tag{11}$$

Note that the region given in (11) is derived from the formula of the asymmetric capacity (8), which corresponds to the case that user Y gets common+private information, whereas user Z gets only the common information. This is emphasized by the superscript y in (11). We have

Lemma 1 For any pair (U, X),

$$\mathcal{R}_i^y(U, X) \subseteq \mathcal{C}_{ID}.$$

Proof We will make use of the idea of common experiment of [3]. Let $\mathcal{C}_n$ be an (n, M_y, M_0, λ) transmission code for the broadcast channel with degraded message sets, with index sets $\mathcal{M}_y$ and $\mathcal{M}_0$. Denote the (asymmetric) rate pair of this code by (R_y, R_0), and the encoder by f. The region of achievable transmission rate pairs is given by (8). Hence the lemma will be proved once we show that, for the rate pair (R_y, R_0) of the transmission code $\mathcal{C}_n$ and any $\gamma > 0$, we can construct an ID code of rate pair (R_y^{ID}, R_z^{ID}) satisfying

$$R_y^{ID} = R_y - R_0 - \gamma \tag{12}$$

$$R_z^{ID} = R_0 - \gamma \tag{13}$$

for all sufficiently large n.

Fix $\theta > 0$ arbitrarily small. Let $\mathcal{C}_{n\theta}$ be an $(n\theta, M_y^\theta, M_0^\theta, \lambda_\theta)$ transmission code for the broadcast channel with degraded message sets, with index sets $\mathcal{M}_y^\theta$ and $\mathcal{M}_0^\theta$. Denote its encoder by f^θ, and the corresponding decoding sets by $\mathcal{D}_{i,k}^{\theta,y}$ and $\mathcal{D}_k^{\theta,z}$. The script θ is added here to distinguish this code from the longer one, $\mathcal{C}_n$. Since θ is arbitrarily small, we can set

$$M_y^\theta = M_0^\theta = \exp n\theta. \tag{14}$$

For the purpose of ID coding, we will construct two sets of mappings, $\{T_{y,i},\ \ i = 1, 2, \ldots, N_y\}$, and $\{T_{z,j},\ \ j = 1, 2, \ldots, N_z\}$, where

$$T_{y,i} : \ \mathcal{M}_y \times \mathcal{M}_0 \ \longrightarrow \ \mathcal{M}_y^\theta, \tag{15}$$

$$T_{z,j} : \quad \mathcal{M}_0 \ \longrightarrow \ \mathcal{M}_0^\theta. \tag{16}$$

An ID code for the broadcast channel can be constructed with the aid of such mappings, as follows. Let $Q_{\mathcal{M}_y\mathcal{M}_0}$ be a uniform distribution on the product set $\mathcal{M}_y \times \mathcal{M}_0$

$$Q_{\mathcal{M}_y\mathcal{M}_0}(r, k) = (M_y M_0)^{-1} \quad \forall\ r \in \mathcal{M}_y,\ \ k \in \mathcal{M}_0.$$

We now specify the encoding and decoding schemes.

Encoding: Random encoding of a pair (i, j) of identification indices is done according to the following steps:

1. Choose randomly a pair of indices $(r, k) \in \mathcal{M}_y \times \mathcal{M}_0$, according to the distribution $Q_{\mathcal{M}_y \mathcal{M}_0}$.
2. Send the codeword $f(r, k)$ via the channel.
3. Send via the channel the codeword $f^\theta(T_{y,i}(r, k), T_{z,j}(k))$.

For convenience, let us use the notation

$$F_{i,j}(r, k) \stackrel{\triangle}{=} f(r, k) f^\theta(T_{y,i}(r, k), T_{z,j}(k)). \tag{17}$$

Note that the encoding scheme makes use of $n(1 + \theta)$ input letters, where θ is arbitrarily small. The random encoder $Q_{i,j}(\cdot)$ resulting from the above three steps is

$$Q_{i,j}(x^n \tilde{x}^{n\theta}) = \sum_{\substack{(r,k): \\ f(r,k)=x^n \\ f^\theta(T_{y,i}(r,k),T_{z,j}(k))=\tilde{x}^{n\theta}}} Q_{\mathcal{M}_y \mathcal{M}_0}(r, k). \tag{18}$$

Decoding: We need only to specify the decoding sets. These are constructed in the natural way, via the corresponding decoding sets of the transmission codes. Thus, the decoding sets for channel W_z are

$$\mathcal{B}_j^z = \bigcup_{k=1}^{M_0} \mathcal{D}_k^z \times \mathcal{D}_{T_{z,j}(k)}^{\theta,z}, \tag{19}$$

and for W_y

$$\mathcal{B}_i^y = \bigcup_{r=1}^{M_y} \bigcup_{k=1}^{M_0} \mathcal{D}_{r,k}^y \times \hat{\mathcal{D}}_{T_{y,i}(r,k)}^{\theta,y}, \tag{20}$$

where

$$\hat{\mathcal{D}}_{T_{y,i}(r,k)}^{\theta,y} \stackrel{\triangle}{=} \bigcup_{k'=1}^{M_0^\theta} \mathcal{D}_{T_{y,i}(r,k),k'}^{\theta,y}. \tag{21}$$

The rate of the resulting ID code is determined solely by the size of the collection of maps $\{T_{y,i}\}_{i=1}^{N_y}$ and $\{T_{z,j}\}_{j=1}^{N_z}$. These two collections of maps can be constructed independently of each other.

This can be justified as follows. Assume we are given the transmission codes $\mathcal{C}_n$ and $\mathcal{C}_{n\theta}$. Let $\{T_{y,i}\}_{i=1}^{N_y}$ and $\{T_{z,j}\}_{j=1}^{N_z}$ be two collections of maps, and construct an ID code as described above. A consequence of the ID code construction is that the probabilities of error of first and second kind for channel Y are independent of the specific maps $\{T_{z,j}\}_{j=1}^{N_z}$. They depends only on the properties of the collection $\{T_{y,i}\}_{i=1}^{N_y}$ and the transmission codes $\mathcal{C}_n$ and $\mathcal{C}_{n\theta}$ used to construct the ID code. Similarly, the probabilities of error for channel Z do not depend on the maps $\{T_{y,i}\}_{i=1}^{N_y}$. Therefore, we can prove lower bounds on N_y and N_z separately. The support of each of the maps $\{T_{y,i}\}_{i=1}^{N_y}$ is of size $\exp(nR_y + nR_0)$, and that of the maps $\{T_{z,j}\}_{j=1}^{N_z}$ is of size $\exp(nR_0)$. Thus, following the lines of [3], we show below, by random selection methods, that for every $\gamma > 0$ and sufficiently large n, we can construct up to

$$N_y = \exp\exp(nR_y + nR_0 - n\gamma) = \exp(M_y M_0 e^{-n\gamma}) \tag{22}$$

and

$$N_z = \exp\exp(nR_0 - n\gamma) = \exp(M_0 e^{-n\gamma}) \tag{23}$$

such maps, and still get probabilities of error of first and second kind, λ_1 and λ_2, arbitrarily small. Note that (22), (23) imply (12), (13).

We proceed now to the construction of these maps, by random selection methods. For $i \in \mathcal{N}_y$, $j \in \mathcal{N}_z$, $r \in \mathcal{M}_y$, and $k \in \mathcal{M}_0$, let $\bar{T}_{y,i}(r,k)$ and $\bar{T}_{z,j}(k)$ be independent random variables uniformly distributed on $\mathcal{M}_y^\theta$ and $\mathcal{M}_0^\theta$, respectively. Let $T_{y,i}(r,k)$ and $T_{z,j}(k)$ be specific realizations of these random variables. We use this convention also for the encoder and decoding sets. Thus, $\bar{F}_{i,j}(r,k)$, $\bar{Q}_{i,j}(x^n x^{n\theta})$, $\bar{\mathcal{B}}_j^z$, and $\bar{\mathcal{B}}_i^y$ stand for the encoder and decoding sets of (17), (18), (19), and (20), respectively, where the maps are random, and the notation without the bar for specific realizations.

Next, (19) and (20) are used to upper bound the probabilities of error of the first kind. Starting with the channel W_y, we have

$$\begin{aligned}
&\sum_{r\in\mathcal{M}_y}\sum_{k\in\mathcal{M}_0} Q_{\mathcal{M}_y\mathcal{M}_0}(r,k) W_y^{n(1+\theta)}\left((\mathcal{B}_i^y)^c \,|F_{i,j}(r,k)\right) \\
&\overset{(a)}{\leq} \frac{1}{M_y M_0}\sum_{r\in\mathcal{M}_y}\sum_{k\in\mathcal{M}_0} W_y^{n(1+\theta)}\left(\left(\mathcal{D}_{r,k}^y \times \hat{\mathcal{D}}_{T_{y,i}(r,k)}^{\theta,y}\right)^c |F_{i,j}(r,k)\right) \\
&\overset{(b)}{\leq} \frac{1}{M_y M_0}\sum_{r\in\mathcal{M}_y}\sum_{k\in\mathcal{M}_0} W_y^{n}\left(\left(\mathcal{D}_{r,k}^y\right)^c |f(r,k)\right)
\end{aligned}$$

$$+\frac{1}{M_y M_0}\sum_{r\in\mathcal{M}_y}\sum_{k\in\mathcal{M}_0} W_y^{n\theta}\left(\left\{\bigcup_{k'=1}^{M_0^\theta}\mathcal{D}^{\theta,y}_{T_{y,i}(r,k),k'}\right\}^c\left|f^\theta(T_{y,i}(r,k),T_{z,j}(k))\right.\right)$$

$$\leq \quad \lambda+\lambda_\theta \qquad \forall(i,j), \tag{24}$$

where (a) follows from (20), and (b) from the coding scheme and the fact that the channel is memoryless. For the second channel we have

$$\begin{aligned}
&\sum_{r\in\mathcal{M}_y}\sum_{k\in\mathcal{M}_0} Q_{\mathcal{M}_y\mathcal{M}_0}(r,k)W_z^{n(1+\theta)}\left(\left(\mathcal{B}_j^z\right)^c|F_{i,j}(r,k)\right)\\
&\overset{(a)}{\leq} \quad \frac{1}{M_yM_0}\sum_{r\in\mathcal{M}_y}\sum_{k\in\mathcal{M}_0} W_z^{n(1+\theta)}\left(\left(\mathcal{D}_k^z\times\mathcal{D}^{\theta,z}_{T_{z,j}(k)}\right)^c|F_{i,j}(r,k)\right)\\
&\leq \quad \frac{1}{M_yM_0}\sum_{r\in\mathcal{M}_y}\sum_{k\in\mathcal{M}_0} W_z^{n}\left((\mathcal{D}_k^z)^c|f(r,k)\right)\\
&\qquad +\frac{1}{M_yM_0}\sum_{r\in\mathcal{M}_y}\sum_{k\in\mathcal{M}_0} W_z^{n\theta}\left(\left(\mathcal{D}^{\theta,z}_{T_{z,j}(k)}\right)^c\left|f^\theta(T_{y,i}(r,k),T_{z,j}(k))\right.\right)\\
&\leq \quad \lambda+\lambda_\theta \qquad \forall(i,j),
\end{aligned} \tag{25}$$

where (a) follows from (19). Therefore, the probability of error of the first kind is upper bounded by $\lambda+\lambda_\theta$ for any choice of mappings $\{T_{y,i}\}_i$ and $\{T_{z,j}\}_j$, due to the code construction. It follows that (22), (23) are achievable with small probability of error of the first kind.

We turn to evaluate the probability of error of the second kind, λ_2. We will use a large-deviational approach, similar to that of [3], to bound the probability of error of second kind for two indices, say 1 and 2, for both channels W_y and W_z. This probability of error depends on the realizations of the mappings, and thus it is a random variable. For fixed realizations of the maps $T_{y,1}$ and $T_{z,1}$, define the random variables

$$\Psi_y(r,k) \quad = \quad \begin{cases} 1 & \text{if } \bar{T}_{y,2}(r,k)=T_{y,1}(r,k)\\ 0 & \text{else,}\end{cases} \tag{26}$$

$$\Psi_z(k) \quad = \quad \begin{cases} 1 & \text{if } \bar{T}_{z,2}(k)=T_{z,1}(k)\\ 0 & \text{else.}\end{cases} \tag{27}$$

Since the random variables $\{\bar{T}_{y,2}(r,k),\ \bar{T}_{z,2}(k)\}_{r,k}$ are independent, $\Psi_y(r,k)$ and $\Psi_z(k)$ are also independent. In addition

$$\mathbb{E}\Psi_y(r,k) \quad = \quad \left(M_y^\theta\right)^{-1} \qquad \forall\ (r,k)\in\mathcal{M}_y\times\mathcal{M}_0 \tag{28}$$

$$\mathbb{E}\Psi_z(k) \;=\; \left(M_0^\theta\right)^{-1} \quad \forall\ k \in \mathcal{M}_0 \tag{29}$$

where $\mathbb{E}$ stands for the expectation operation. Let η_y and η_0 be two parameters satisfying $1/M_y^\theta < \eta_y < 1$ and $1/M_0^\theta < \eta_0 < 1$. A simple calculation shows that

$$\begin{aligned} D\left(\eta_y \middle\| \frac{1}{M_y^\theta}\right) &> n\theta\eta_y - 1, \\ D\left(\eta_0 \middle\| \frac{1}{M_0^\theta}\right) &> n\theta\eta_0 - 1 \end{aligned} \tag{30}$$

where $D(\eta\|\mu)$ stands for the divergence between $(\eta, 1-\eta)$ and $(\mu, 1-\mu)$. Therefore, [2, Lemma LD] implies that

$$P\left[\sum_{r\in\mathcal{M}_y}\sum_{k\in\mathcal{M}_0}\Psi_y(r,k) > \eta_y M_y M_0\right] \le \exp\left[-M_y M_0 D\left(\eta_y\|\frac{1}{M_y^\theta}\right)\right] \le \exp\left(-M_y M_0(n\theta\eta_y - 1)\right) \tag{31}$$

and similarly

$$P\left[\sum_{k\in\mathcal{M}_0}\Psi_z(k) > \eta_0 M_0\right] \le \exp\left[-M_0 D\left(\eta_0\|\frac{1}{M_0^\theta}\right)\right] \le \exp\left(-M_0(n\theta\eta_0 - 1)\right). \tag{32}$$

For simplicity of notation, let us fix the realizations $T_{y,2}(\cdot,\cdot)$, $T_{z,2}(\cdot)$. We will resume to the notation $\bar{T}_{y,2}(\cdot,\cdot)$, $\bar{T}_{z,2}(\cdot)$ when the randomness comes into account. The probability of error of the second kind for the channel W_y can be bounded as

$$\begin{aligned} Q_{1,j}W_y^{n(1+\theta)}(\mathcal{B}_2^y) &= \sum_{(r,k)\in\mathcal{M}_y\times\mathcal{M}_0} Q_{\mathcal{M}_y\mathcal{M}_0}(r,k)W^{n(1+\theta)}\left(\mathcal{B}_2^y\|F_{1,j}(r,k)\right) \\ &\le \sum_{(r,k)\in\mathcal{M}_y\times\mathcal{M}_0} Q_{\mathcal{M}_y\mathcal{M}_0}(r,k)W^{n(1+\theta)}\left((\mathcal{B}_1^y)^c\|F_{1,j}(r,k)\right) \\ &\quad + \sum_{(r,k)\in\mathcal{M}_y\times\mathcal{M}_0} Q_{\mathcal{M}_y\mathcal{M}_0}(r,k)W^{n(1+\theta)}\left(\mathcal{B}_1^y\cap\mathcal{B}_2^y\|F_{1,j}(r,k)\right) \\ &= L_1 + L_2, \end{aligned} \tag{33}$$

where L_1 stands for the first sum in the far right hand side of (33), and L_2 stands for the second sum. By definition of the sets $\mathcal{B}_i^y$, L_1 satisfies:

$$L_1 \le \lambda + \lambda_\theta. \tag{34}$$

To evaluate L_2, notice that since $\{\mathcal{D}_{r,k}^y\}_{r,k}$ is a disjoint family of sets, we have

$$\mathcal{B}_1^y\cap\mathcal{B}_2^y \;=\; \left\{\bigcup_{r=1}^{M_y}\bigcup_{k=1}^{M_0}\mathcal{D}_{r,k}^y\times\hat{\mathcal{D}}_{T_{y,1}(r,k)}^{\theta,y}\right\}\bigcap\left\{\bigcup_{r=1}^{M_y}\bigcup_{k=1}^{M_0}\mathcal{D}_{r,k}^y\times\hat{\mathcal{D}}_{T_{y,2}(r,k)}^{\theta,y}\right\}$$

$$= \bigcup_{r=1}^{M_y} \bigcup_{k=1}^{M_0} \mathcal{D}^y_{r,k} \times \left[\hat{\mathcal{D}}^{\theta,y}_{T_{y,1}(r,k)} \bigcap \hat{\mathcal{D}}^{\theta,y}_{T_{y,2}(r,k)}\right], \tag{35}$$

and therefore L_2 can be further bounded as

$$\begin{aligned}
L_2 &\leq \frac{1}{M_y M_0} \sum_{(r,k)\in\mathcal{M}_y\times\mathcal{M}_0} W_y^{n(1+\theta)} \left(\bigcup_{r'=1}^{M_y} \bigcup_{k'=1}^{M_0} \mathcal{D}^y_{r',k'} \times \left[\hat{\mathcal{D}}^{\theta,y}_{T_{y,1}(r',k')} \bigcap \hat{\mathcal{D}}^{\theta,y}_{T_{y,2}(r',k')}\right] |F_{1,j}(r,k) \right) \\
&\leq \frac{1}{M_y M_0} \sum_{(r,k)\in\mathcal{M}_y\times\mathcal{M}_0} W_y^{n(1+\theta)} \left(\bigcup_{r'\neq r} \bigcup_{k'\neq k} \mathcal{D}^y_{r',k'} \times \left[\hat{\mathcal{D}}^{\theta,y}_{T_{y,1}(r',k')} \bigcap \hat{\mathcal{D}}^{\theta,y}_{T_{y,2}(r',k')}\right] |F_{1,j}(r,k) \right) \\
&\quad + \frac{1}{M_y M_0} \sum_{(r,k)\in\mathcal{M}_y\times\mathcal{M}_0} W_y^{n(1+\theta)} \left(\mathcal{D}^y_{r,k} \times \left[\hat{\mathcal{D}}^{\theta,y}_{T_{y,1}(r,k)} \bigcap \hat{\mathcal{D}}^{\theta,y}_{T_{y,2}(r,k)}\right] |F_{1,j}(r,k) \right) \\
&\leq \lambda + \frac{1}{M_y M_0} \sum_{(r,k)\in\mathcal{M}_y\times\mathcal{M}_0} W_y^{n(1+\theta)} \left(\mathcal{D}^y_{r,k} \times \left[\hat{\mathcal{D}}^{\theta,y}_{T_{y,1}(r,k)} \bigcap \hat{\mathcal{D}}^{\theta,y}_{T_{y,2}(r,k)}\right] |F_{1,j}(r,k) \right) \\
&\overset{(a)}{=} \lambda + \frac{1}{M_y M_0} \sum_{\substack{(r,k)\in\mathcal{M}_y\times\mathcal{M}_0 \\ T_{y,1}(r,k)=T_{y,2}(r,k)}} W_y^{n(1+\theta)} \left(\mathcal{D}^y_{r,k} \times \left[\hat{\mathcal{D}}^{\theta,y}_{T_{y,1}(r,k)} \bigcap \hat{\mathcal{D}}^{\theta,y}_{T_{y,2}(r,k)}\right] |F_{1,j}(r,k) \right) \\
&\leq \lambda + \frac{1}{M_y M_0} \sum_{(r,k)\in\mathcal{M}_y\times\mathcal{M}_0} \mathbb{I}\left\{T_{y,1}(r,k) = T_{y,2}(r,k)\right\},
\end{aligned} \tag{36}$$

where $\mathbb{I}\{\cdot\}$ stands for the indicator function, and in (a) we do not have to sum over (r,k) for which $T_{y,1}(r,k) \neq T_{y,2}(r,k)$, since in that case $\hat{\mathcal{D}}^{\theta,y}_{T_{y,1}(r,k)} \cap \hat{\mathcal{D}}^{\theta,y}_{T_{y,2}(r,k)}$ is empty.

Inequality (36) holds for any realization of the mappings $T_{y,1}(\cdot,\cdot)$, $T_{y,2}(\cdot,\cdot)$. We now resume to the random notation. Recall that we have fixed a realization $T_{y,1}$, and that $\bar{T}_{y,2}$ is random. Thus we have

$$\frac{1}{M_y M_0} \sum_{(r,k)\in\mathcal{M}_y\times\mathcal{M}_0} \mathbb{I}\left\{T_{y,1}(r,k) = \bar{T}_{y,2}(r,k)\right\} = \frac{1}{M_y M_0} \sum_{(r,k)\in\mathcal{M}_y\times\mathcal{M}_0} \Psi_y(r,k). \tag{37}$$

Collecting (33), (34), (36), and (37), we have that for the random maps $\bar{T}_{y,2}$,

$$Q_{1,j} W_y^{n(1+\theta)}(\mathcal{B}_2^y) \leq 2\lambda + \lambda_\theta + \frac{1}{M_y M_0} \sum_{(r,k)\in\mathcal{M}_y\times\mathcal{M}_0} \Psi_y(r,k), \quad \forall j. \tag{38}$$

Similarly, using the same arguments, we show that

$$Q_{2,j} W_y^{n(1+\theta)}(\mathcal{B}_1^y) \leq 2\lambda + \lambda_\theta + \frac{1}{M_y M_0} \sum_{(r,k)\in\mathcal{M}_y\times\mathcal{M}_0} \Psi_y(r,k), \quad \forall j. \tag{39}$$

Fix $\eta_y > 0$. By (31), (38), and (39), we have

$$P\left(Q_{1,j}W_y^{n(1+\theta)}(\mathcal{B}_2^y) > 2\lambda + \lambda_\theta + \eta_y\right) \leq \exp\left(-M_y M_0(n\theta\eta_y - 1)\right), \tag{40}$$

$$P\left(Q_{2,j}W_y^{n(1+\theta)}(\mathcal{B}_1^y) > 2\lambda + \lambda_\theta + \eta_y\right) \leq \exp\left(-M_y M_0(n\theta\eta_y - 1)\right). \tag{41}$$

The right hand sides of (40), (41), do not depend on the specific realizations of $T_{y,1}$, and thus the bounds hold for the random $\bar{T}_{y,1}$ as well. We now repeat these steps for $i = 3, 4, \ldots N_y$. By these steps we consider N_y^2 pairs $Q_{i,j}, Q_{i',j}$. Hence, by the union bound, a code with probability of error of second kind $2\lambda + \lambda_\theta + \eta_y$ and of size N_y is guaranteed to exist provided

$$N_y^2 \exp\left(-M_y M_0(n\theta\eta_y - 1)\right) < 1,$$

or

$$N_y < \exp(M_y M_0 \epsilon') \tag{42}$$

for some fixed, sufficiently small ϵ'. This proves that (22) is achievable.

Our last step is to calculate the probability of error of the second kind for the channel W_z. This follows closely the steps of that for the channel W_y. Thus

$$\begin{aligned} Q_{i,1}W_z^{n(1+\theta)}(\mathcal{B}_2^z) &= \sum_{(r,k)\in\mathcal{M}_y\times\mathcal{M}_0} Q_{\mathcal{M}_y\mathcal{M}_0}(r,k)W_z^{n(1+\theta)}\left(\mathcal{B}_2^z|F_{i,1}(r,k)\right) \\ &\leq \sum_{(r,k)\in\mathcal{M}_y\times\mathcal{M}_0} Q_{\mathcal{M}_y\mathcal{M}_0}(r,k)W_z^{n(1+\theta)}\left((\mathcal{B}_1^z)^c|F_{i,1}(r,k)\right) \\ &\quad + \sum_{(r,k)\in\mathcal{M}_y\times\mathcal{M}_0} Q_{\mathcal{M}_y\mathcal{M}_0}(r,k)W_z^{n(1+\theta)}\left(\mathcal{B}_1^z\cap\mathcal{B}_2^z|F_{i,1}(r,k)\right) \\ &= L_3 + L_4, \end{aligned} \tag{43}$$

where L_3 stands for the first sum in the far right hand side of (43), and L_4 for the second sum. By definition of the sets $\mathcal{B}_j^z$, we have

$$L_3 \leq \lambda + \lambda_\theta. \tag{44}$$

Regarding L_4, we have

$$\begin{aligned} \mathcal{B}_1^z \cap \mathcal{B}_2^z &= \left\{\bigcup_{k=1}^{M_0} \mathcal{D}_k^z \times \mathcal{D}_{T_{z,1}(k)}^{\theta,z}\right\} \bigcap \left\{\bigcup_{k=1}^{M_0} \mathcal{D}_k^z \times \mathcal{D}_{T_{z,2}(k)}^{\theta,z}\right\} \\ &= \bigcup_{k=1}^{M_0} \mathcal{D}_k^z \times \left[\mathcal{D}_{T_{z,1}(k)}^{\theta,z} \bigcap \mathcal{D}_{T_{z,2}(k)}^{\theta,z}\right], \end{aligned} \tag{45}$$

so L_4 is bounded by

$$\begin{aligned}
L_4 &\leq \frac{1}{M_y M_0} \sum_{(r,k)\in\mathcal{M}_y\times\mathcal{M}_0} W_z^{n(1+\theta)} \left(\bigcup_{k'=1}^{M_0} \mathcal{D}_{k'}^z \times \left[\mathcal{D}_{T_{z,1}(k')}^{\theta,z} \bigcap \mathcal{D}_{T_{z,2}(k')}^{\theta,z} \right] | F_{i,1}(r,k) \right) \\
&\leq \frac{1}{M_y M_0} \sum_{(r,k)\in\mathcal{M}_y\times\mathcal{M}_0} W_z^{n(1+\theta)} \left(\bigcup_{k'\neq k} \mathcal{D}_{k'}^z \times \left[\mathcal{D}_{T_{z,1}(k')}^{\theta,z} \bigcap \mathcal{D}_{T_{z,2}(k')}^{\theta,z} \right] | F_{i,1}(r,k) \right) \\
&\quad + \frac{1}{M_y M_0} \sum_{(r,k)\in\mathcal{M}_y\times\mathcal{M}_0} W_z^{n(1+\theta)} \left(\mathcal{D}_k^z \times \left[\mathcal{D}_{T_{z,1}(k)}^{\theta,z} \bigcap \mathcal{D}_{T_{z,2}(k)}^{\theta,z} \right] | F_{i,1}(r,k) \right) \\
&\overset{(a)}{=} \lambda + \frac{1}{M_y M_0} \sum_{\substack{(r,k)\in\mathcal{M}_y\times\mathcal{M}_0 \\ T_{z,1}(k)=T_{z,2}(k)}} W_z^{n(1+\theta)} \left(\mathcal{D}_k^z \times \left[\mathcal{D}_{T_{z,1}(k)}^{\theta,z} \bigcap \mathcal{D}_{T_{z,2}(k)}^{\theta,z} \right] | F_{i,1}(r,k) \right) \\
&\leq \lambda + \frac{1}{M_y M_0} \sum_{(r,k)\in\mathcal{M}_y\times\mathcal{M}_0} \mathbb{I}\{T_{z,1}(k) = T_{z,2}(k)\} \\
&= \lambda + \frac{1}{M_0} \sum_{k\in\mathcal{M}_0} \mathbb{I}\{T_{z,1}(k) = T_{z,2}(k)\} \qquad (46)
\end{aligned}$$

where (a) follows from the fact that if $T_{z,1}(k) \neq T_{z,2}(k)$, then $\mathcal{D}_{T_{z,1}(k)}^{\theta,z} \bigcap \mathcal{D}_{T_{z,2}(k)}^{\theta,z}$ is empty. Resuming now to random notation, we have

$$\frac{1}{M_0} \sum_{k\in\mathcal{M}_0} \mathbb{I}\{T_{z,1}(k) = \bar{T}_{z,2}(k)\} = \frac{1}{M_0} \sum_{k\in\mathcal{M}_0} \Psi_z(k). \qquad (47)$$

Collecting (43), (44), (46), (47), we have that for the random maps $\bar{T}_{z,2}$

$$Q_{i,1} W_z^{n(1+\theta)}(\mathcal{B}_2^z) \leq 2\lambda + \lambda_\theta + \frac{1}{M_0} \sum_{k\in\mathcal{M}_0} \Psi_z(k) \quad \forall i, \qquad (48)$$

and similarly

$$Q_{i,2} W_z^{n(1+\theta)}(\mathcal{B}_1^z) \leq 2\lambda + \lambda_\theta + \frac{1}{M_0} \sum_{k\in\mathcal{M}_0} \Psi_z(k) \quad \forall i. \qquad (49)$$

Fix $\eta_0 > 0$. By (32), (48), and (49), we obtain

$$\begin{aligned}
P\left(Q_{i,1} W_z^{n(1+\theta)}(\mathcal{B}_2^z) > 2\lambda + \lambda_\theta + \eta_0 \right) &\leq \exp(-M_0(n\theta\eta_0 - 1)), \\
P\left(Q_{i,2} W_z^{n(1+\theta)}(\mathcal{B}_1^z) > 2\lambda + \lambda_\theta + \eta_0 \right) &\leq \exp(-M_0(n\theta\eta_0 - 1)),
\end{aligned}$$

therefore, by the union bound, a code with probability of error of second kind $2\lambda + \lambda_\theta + \eta_0$ and of size N_z exists provided

$$N_z < \exp(M_0 \epsilon')$$

for some fixed, sufficiently small ϵ'. Therefore (23) is achievable. This concludes the proof of Lemma 1. □

Lemma 1 is proved for the general discrete memoryless broadcast channel. Therefore the roles of Y and Z in (11) can be interchanged, and the resulting region is ID achievable as there is no hierarchy between the Y and Z channels. Thus define

$$\begin{aligned} \mathcal{R}_i^z(U,X) \stackrel{\triangle}{=} \{(R_y,R_z): \quad R_y &\leq I(U;Y), \\ R_z &\leq I(X;Z|U)+I(U;Y) \bigwedge I(X;Z)\}. \end{aligned} \tag{50}$$

By Lemma 1, $\mathcal{R}_i^z(U,X)$ is ID-achievable for any pair (U,X). In the next lemma we use Lemma 1 to prove the direct part of Theorem 1.

Lemma 2 For any input X, the pair (R_y^{ID}, R_z^{ID}) given by

$$\begin{aligned} R_y^{ID} &= I(X;Y) \\ R_z^{ID} &= I(X;Z) \end{aligned} \tag{51}$$

is an achievable ID rate pair.

Proof We use the characterizations (11) and (50). By setting $U \equiv X$ we conclude that the following regions are ID achievable

$$\begin{aligned} \mathcal{R}_i^y(X) \stackrel{\triangle}{=} \{(R_y,R_z): \quad R_y &\leq I(X;Z) \bigwedge I(X;Y), \\ R_z &\leq I(X;Z)\}, \end{aligned} \tag{52}$$

and

$$\begin{aligned} \mathcal{R}_i^z(X) \stackrel{\triangle}{=} \{(R_y,R_z): \quad R_y &\leq I(X;Y), \\ R_z &\leq I(X;Y) \bigwedge I(X;Z)\}. \end{aligned} \tag{53}$$

For a given input distribution X, $I(X;Y)$ and $I(X;Z)$ determine the ID code construction. If $I(X;Y) < I(X;Z)$, we choose the representation (52) and construct an ID code according to the proof of Lemma 1, with $U \equiv X$. If $I(X;Y) \geq I(X;Z)$, we choose the representation (53) and construct an ID code according to the proof of Lemma 1, with $U \equiv X$, and with the roles of Y and Z interchanged. This shows that (51) is achievable. □

Remark. Lemma 2 suggests a rule for determining whether to use the representation (and corresponding code construction) (52) or (53), which is optimal in the sense that it exhausts the capacity region. According to this rule, if $I(X;Z) \leq I(X;Y)$, we construct the ID code via a transmission code which gives to Y only common messages, and to Z common+private messages. Similarly for $I(X;Y) < I(X;Z)$, with Y and Z swipped. This rule is somewhat counter-intuitive. if we examine it from point of view of transmission. To illustrate, let (W_y, W_z) be a degraded broadcast channel, with Z being the degraded component. Then any transmission message sent to Z can be decoded also by Y, and therefore, borrowing intuition from the classical (Shannon transmission) model, we would use the message set of Z as a common experiment. Note, however, that the preferred construction is exactly the opposite. Since Z is the degraded component, we have $I(X;Z) \leq I(X;Y)$, and therefore by the proof of Lemma 2, we should use a transmission code for the broadcast channel with degraded message set, where Y gets only common rate, and Z gets common+private.

The next lemma, stated and proved in [9], is the basis for the proof of the converse result.

Lemma 3 Distribution Packing [9, Lemma 6]: Let W^n be an arbitrary channel. Assume that for any $n > n_0$ there exists a collection of distributions $\mathcal{F} = \{Q_i\}_{i=1}^{N}$ such that

$$\frac{1}{n} \log \log N \geq R$$

and

$$\min_{i \neq j} d(Q_i W^n, Q_j W^n) \geq 2(1-\mu).$$

Then for every $\gamma < (\mu/4)\min(1, R)$ there exists a subcollection $\mathcal{P}$ of $\mathcal{F}$ such that

$$|\mathcal{P}| \geq \exp\exp(nR) - \exp\exp(nR - n\gamma)$$

and

$$\frac{1}{n} I(Q_i; Q_i W^n) \geq R(1-4\mu)$$

for every $Q_i \in \mathcal{P}$ and $n > n'$, where n' depends only on the alphabets sizes and on the parameters μ, γ.

In the next lemma, we use the distribution packing result to give an outer bound on the capacity region.

Lemma 4 Fix $\rho > \lambda_1 + \lambda_2$. For any broadcast channel

$$C_{ID}(\lambda_1, \lambda_2) \subseteq \text{closure}(\liminf_{n\to\infty} \frac{1}{n} C_n(\rho)), \tag{54}$$

where

$$\begin{aligned} C_n(\rho) = \bigcup_{X^n} \{(R_y, R_z) : \quad & R_y(1-4\rho) \leq I(X^n; Y^n) \\ & R_z(1-4\rho) \leq I(X^n; Z^n)\}, \end{aligned}$$

the union is over all input distributions X^n, and (Y^n, Z^n) are outputs of the broadcast channel due to input X^n

Proof: Let (R_y, R_z) be (λ_1, λ_2)- ID achievable pair. Set $\rho > \lambda_1 + \lambda_2$. To prove the lemma it suffices to show that for all sufficiently large n we can find an ID code $\mathcal{Q}_n$ and a distribution $Q_{i,j} \in \mathcal{Q}_n$ such that

$$R_y(1-4\rho) \leq \frac{1}{n} I(Q_{i,j}; Q_{i,j} W_y^n) \tag{55}$$

and

$$R_z(1-4\rho) \leq \frac{1}{n} I(Q_{i,j}; Q_{i,j} W_z^n). \tag{56}$$

By the achievability of (R_y, R_z), for all sufficiently large n we have collection of distributions

$$\mathcal{Q}_n = \{Q_{i,j}, i = 1 \ldots \exp\exp nR_y, j = 1 \ldots \exp\exp nR_z\} \tag{57}$$

such that (4), (5) hold. We use the usual notation for the sets of indices

$$\mathcal{N}_y = \{1, 2, \ldots \exp\exp nR_y\},$$

$$\mathcal{N}_z = \{1, 2, \ldots \exp\exp nR_z\},$$

and define

$$\mathcal{N} \stackrel{\Delta}{=} \mathcal{N}_y \times \mathcal{N}_z.$$

Without loss of generality assume $R_z < R_y$. In particular , let $R_z = R_y - \mu$. For any index $j \in \mathcal{N}_z$ we have

$$d(Q_{i,j} W_y^n, Q_{k,j} W_y^n) \geq 2(1-\rho), \quad \forall\, k \neq i. \tag{58}$$

Choose γ sufficiently small so that

$$\gamma < \frac{\rho}{4}\min(1, R_y). \tag{59}$$

By (58) and Lemma 3 there exist subcollections of indices

$$\mathcal{I}_y(j) \subseteq \mathcal{N}_y \quad \forall j \in \mathcal{N}_z \tag{60}$$

such that

$$|\mathcal{I}_y(j)| \geq \exp\exp nR_y - \exp\exp n(R_y - \gamma) \tag{61}$$

and

$$\frac{1}{n} I(Q_{i,j}; Q_{i,j}W_y^n) \geq R_y(1 - 4\rho) \qquad \forall\, i \in \mathcal{I}_y(j) \tag{62}$$

for all sufficiently large n depending on γ, ρ. We claim that

$$\bigcap_{j=1}^{\exp\exp nR_z} \mathcal{I}_y(j) \neq \emptyset. \tag{63}$$

Indeed, it can be verified that:

$$\Big|\bigcap_{j=1}^{\exp\exp nR_z} \mathcal{I}_y(j)\Big| \geq \exp\exp nR_y - \exp\exp n(R_y - \gamma) \cdot \exp\exp nR_z \tag{64}$$

which is positive for sufficiently large n. Therefore there exists at least one index $i^* \in \mathcal{N}_y$ such that:

$$\frac{1}{n} I(Q_{i^*,j}; Q_{i^*,j}W_y^n) \geq R_y(1 - 4\rho) \qquad \forall\ j \in \mathcal{N}_z. \tag{65}$$

Now fix the message index of user Y to be i^*, and apply again Lemma 3 for the single user channel W_z^n. Since

$$d(Q_{i^*,j}W_z^n, Q_{i^*,l}W_z^n) \geq 2(1 - \rho) \quad \forall j \neq l \tag{66}$$

there exists a subcollection

$$\mathcal{I}_z(i^*) \subseteq \mathcal{N}_z \tag{67}$$

such that

$$|\mathcal{P}(i^*)| \geq \exp\exp nR_z - \exp\exp n(R_z - \gamma) \tag{68}$$

and

$$\frac{1}{n} I(Q_{i^*,j}; Q_{i^*,j}W_z^n) \geq R_z(1 - 4\rho) \qquad \forall j \in \mathcal{I}_z(i^*). \tag{69}$$

From (65) and (69) we conclude that one can find a distribution $Q_{i,j} \in \mathcal{Q}_n$ such that

$$\begin{aligned} R_y(1-4\rho) &\leq \frac{1}{n} I(Q_{i,j}; Q_{i,j} W_y^n), \\ R_z(1-4\rho) &\leq \frac{1}{n} I(Q_{i,j}; Q_{i,j} W_z^n), \end{aligned}$$

which proves the lemma. □

Observe that in the proof of Lemma 4 we did not make use of the fact that the channel is memoryless. Hence the outer bound of Lemma 4 holds for any channel. We are now ready for the proof of our main result, Theorem 1.

Proof of Theorem 1. The direct part follows immediately from Lemma 2. To prove the converse, we have to pass to single letter expressions in Lemma 4, and then take the limit of small probabilities of error. Therefore, using the memorylessness of the channel, we can write for the outputs Y, Z of the broadcast channel

$$\begin{aligned} \frac{1}{n} I(X^n; Y^n) &\leq \frac{1}{n} \sum_{i=1}^{n} I(X_i; Y_i) \\ &\leq I(\bar{X}; \bar{Y}). \end{aligned} \tag{70}$$

$$\begin{aligned} \frac{1}{n} I(X^n; Z^n) &\leq \frac{1}{n} \sum_{i=1}^{n} I(X_i; Z_i) \\ &\leq I(\bar{X}; \bar{Z}), \end{aligned} \tag{71}$$

where

$$P_{\bar{X}} = \frac{1}{n} \sum_{i=1}^{n} P_{X_i},$$

$\bar{Y}$ (resp. $\bar{Z}$) is the output of W_y (resp. W_z) due to input $\bar{X}$, and the second inequality in (70) and (71) is due to the concavity of the mutual information in the input distribution. Combining Lemma 4 and (70), (71), we have

$$\mathcal{C}_{ID}(\lambda_1, \lambda_2) \subseteq \bigcup_X \{(R_y, R_z) : \quad R_y(1-4\rho) \leq I(X;Y) \tag{72}$$

$$R_z(1-4\rho) \leq I(X;Z)\}. \tag{73}$$

The converse now follows by taking the limit $\lambda_1 + \lambda_2 \to 0$, and $\rho \to 0$. This completes the proof of Theorem 1. □

References

[1] Abbas A. El Gamal, "The capacity of a class of broadcast channels," *IEEE Trans. Inform. Theory*, vol. 25, pp. 166–169, March 1979.

[2] R. Ahlswede and G. Dueck, "Identification via channels," *IEEE Trans. Inform. Theory*, vol. 35, pp. 15–29, Jan. 1989

[3] —, "Identification in the presence of feedback-A discovery of new capacity formulas," *IEEE Trans. Inform. Theory*, vol. 35, pp 30–36, Jan. 1989

[4] R. Ahlswede and J. Körner, "Source coding with side information and a converse for degraded broadcast channels," *IEEE Trans. Inform. Theory*,vol. 21, pp. 629–637, Nov. 1975.

[5] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, John Wiley Inc. N.Y.

[6] I. Csiszár, J. Körner, *Information Theory. Coding Theorems for Discrete Memoryless Systems*, Academic press, London, 1981.

[7] T. S. Han, S. Verdú, "New results in the theory of identification via channels," *IEEE Trans. Inform. Theory*, vol. 38, pp. 14–25, 1992

[8] J. Körner and K. Marton, "General broadcast channel with degraded message sets," *IEEE Trans. Inform. Theory*, vol. 23, pp. 60–64, Jan. 1977.

[9] Y. Steinberg, "New converses in the theory of identification via channels," *IEEE Trans. Inform. Theory*, vol. 44, pp. 984–998, May 1998.

[10] B. Verboven and E. van der Meulen, "Capacity bounds for identification via broadcast channels that are optimal for the deterministic broadcast channel," *IEEE Trans. Inform. Theory*, vol. 36, pp. 1197–1205, Nov. 1990

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